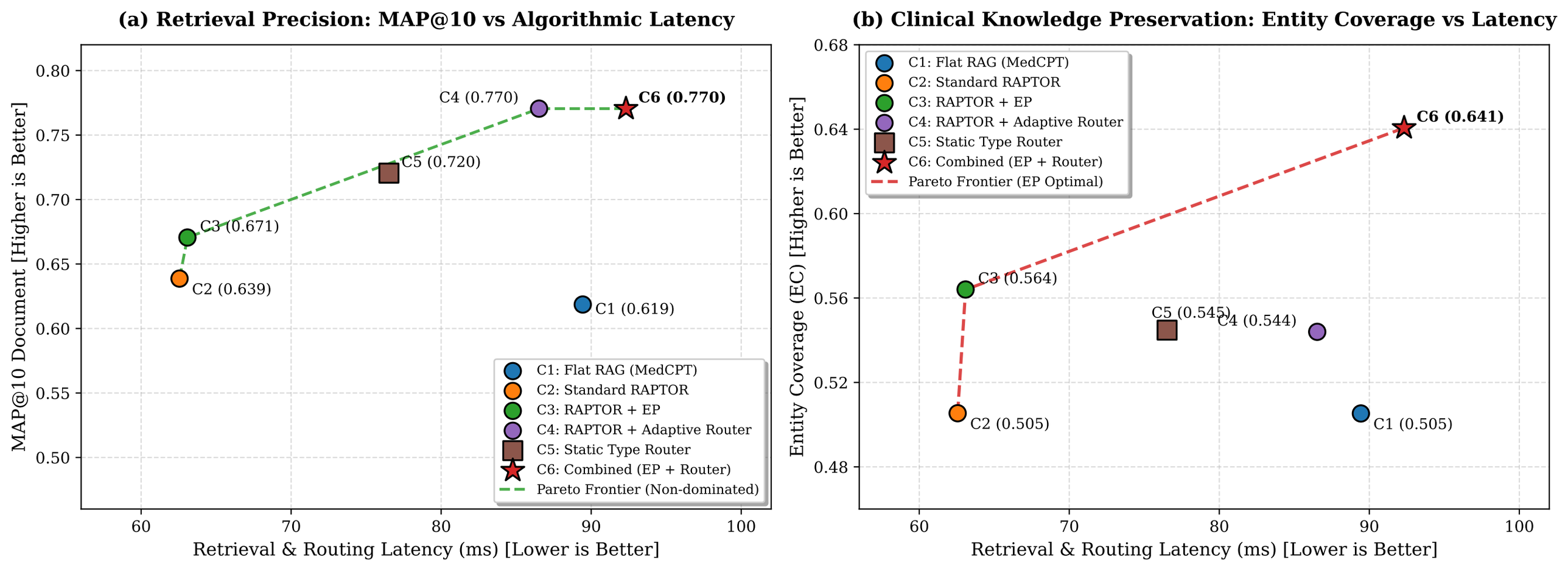
(a) Retrieval Precision: MAP@10 vs Algorithmic Latency
MAP@10 Document [Higher is Better]
Retrieval & Routing Latency (ms) [Lower is Better]
0.80
0.75
0.70
0.65
0.60
0.55
0.50
60
70
80
90
100
C4 (0.770)
C6 (0.770)
C5 (0.720)
C3 (0.671)
C2 (0.639)
C1 (0.619)
C1: Flat RAG (MedCPT)
C2: Standard RAPTOR
C3: RAPTOR + EP
C4: RAPTOR + Adaptive Router
C5: Static Type Router
C6: Combined (EP + Router)
Pareto Frontier (Non-dominated)
(b) Clinical Knowledge Preservation: Entity Coverage vs Latency
Entity Coverage (EC) [Higher is Better]
Retrieval & Routing Latency (ms) [Lower is Better]
0.68
0.64
0.60
0.56
0.52
0.48
60
70
80
90
100
C6 (0.641)
C3 (0.564)
C5 (0.545)
C4 (0.544)
C2 (0.505)
C1 (0.505)
C1: Flat RAG (MedCPT)
C2: Standard RAPTOR
C3: RAPTOR + EP
C4: RAPTOR + Adaptive Router
C5: Static Type Router
C6: Combined (EP + Router)
Pareto Frontier (EP Optimal)